Front zone



File: Front.kicad_sch

Rear zone



File: Rear.kicad_sch

Cockpit zone



File: Cockpit.kicad_sch

Power



File: Power.kicad_sch

Proposed circuits: six CAN front ends, three analog current sensors and three buck converters.
Open: bare-MCU/actuator integration, final protection and power validation, hierarchy, layout.
Local labels do not connect across sheets; finish hierarchy before board import.
See Design-review.csv and the shared Lab 4 Writeup for decisions and unresolved checks.

PRELIMINARY; partial circuits, not fabrication-ready

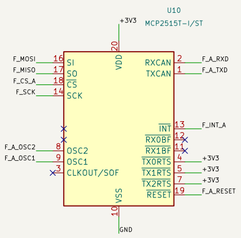
Sheet: /
File: Lab4.kicad_sch

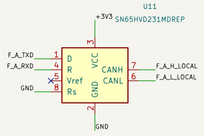
Title: Lab 4 – Three-zone PCB

Size: A4	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. 10.0.4		Id: 1/5

Front zone — CAN and servo current sensing

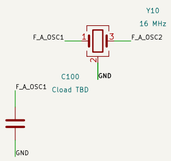
CAN A: 16 MHz / 500 kbit/s / 3.3 V













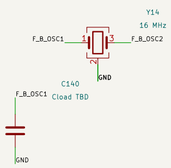






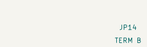
RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V









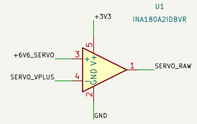


RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

HOST INTERFACE (to be integrated with the bare MCU / Pi header)
SPI1: PA5=SCK, PA6=MISO, PA7=MOSI; PB6=CS_A; PA10=INT_A; PC7=CS_B; PB5=INT_B.
These named nets are LOCAL to this zone. All +3V3 / +5V / GND / CAN backbone nets need hierarchy ports before full-board integration.

SERVO SHUNT — calculated proposal (not measured)













6.6 V regulator -> 10 mOhm shunt -> servo +
Servo GND returns directly to power ground.
Kelvin sense: separate thin traces from shunt pads to IN+/IN-.
Gain 50: Vout = 0.5 V/A. Proposed range 0-5 A -> 0-2.5 V.
12-bit / 3.3 V: 1.61 mA/count (ideal quantization).
At 5 A: shunt drop 50 mV; loss 0.25 W. Select >=1 W.
RC: 4.7k + 100n -> 339 Hz; proposed ADC 4 ksample/s.
Near-zero amplifier output clips; calibrate offset and gain.
This unidirectional circuit cannot measure regenerated current.
No servo idle/stall/noise measurements yet.
5 A measurement range is NOT a 5 A driver or regulator rating.

MCU, PWM, LEDs and SWD integration remains to be completed.
PRELIMINARY — incomplete board; not for fabrication

Team 4

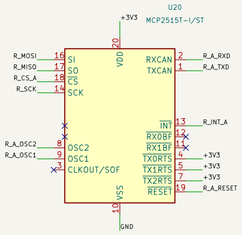
Sheet: /Front zone/
File: Front.kicad_sch

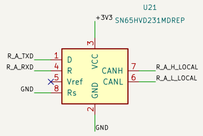
Title: Front zone — CAN and servo current sensing

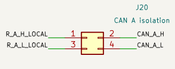
Size: A2	Date: 2026-10-18	**Rev: 0.1**
KiCad E.D.A. 10.0.4		Id: 2/5

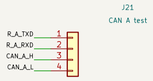
Rear zone — CAN and motor current sensing

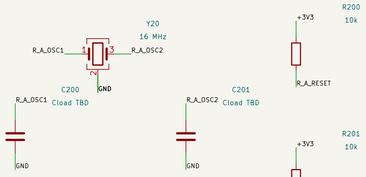
CAN A: 16 MHz / 500 kbit/s / 3.3 V













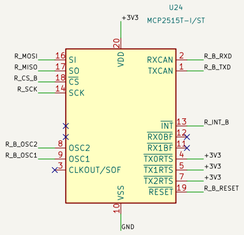


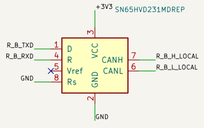


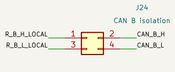


RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V









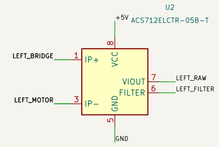




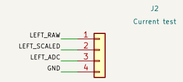
RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

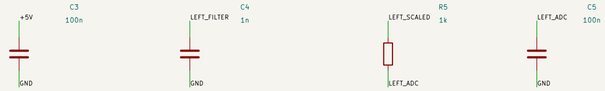
HOST INTERFACE (to be integrated with the bare MCU / Pi header)
SPI1: PA5=SCK, PA6=MISO, PA7=MOSI; PB6=CS_A; PA10=INT_A; PC7=CS_B; PB5=INT_B.
These named nets are LOCAL to this zone. All +3V3 / +5V / GND / CAN backbone nets need hierarchy ports before full-board integration.

LEFT MOTOR: L298 output -> ACS712 -> motor lead

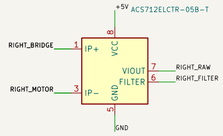




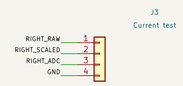


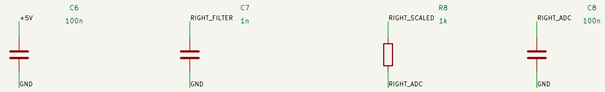


RIGHT MOTOR: L298 output -> ACS712 -> motor lead









Each motor: +/-5 A: VADC = 1.25 V + 0.0925 V/A * I.
Ideal 12-bit scale: 8.71 mA/count at VREF=3.3 V.
10k/10k divider keeps a 5 V sensor output at 2.5 V.
Filter source resistance: (10k || 10k) + 1k = 6k.
6k * 100n -> 265 Hz. Proposed ADC: 4 ksample/s/channel.
Stop motors before zero calibration; preserve signed readings.
Hall sensor has internal conductor, not an external shunt.
At 5 A and typical 1.2 mOhm, conductor loss = 30 mW.
Motor return lead connects to the OTHER L298 output, not GND.

MCU, L298N, flyback, motor/encoder headers remain to be completed.
PRELIMINARY — incomplete board; not for fabrication

Team 4

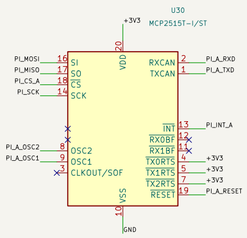
Sheet: /Rear zone/
File: Rear.kicad_sch

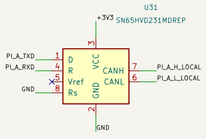
Title: Rear zone — CAN and motor current sensing

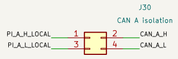
Size: A2	Date: 2026-10-18	**Rev: 0.1**
KiCad E.D.A. 10.0.4		Id: 3/5

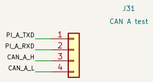
Cockpit zone — dual CAN

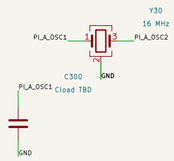
CAN A: 16 MHz / 500 kbit/s / 3.3 V









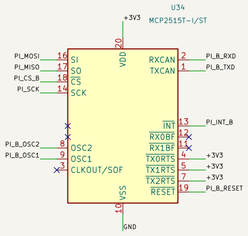


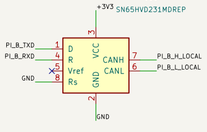


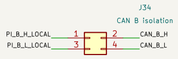


RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

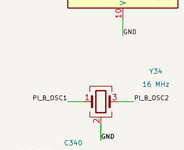
CAN B: 16 MHz / 500 kbit/s / 3.3 V















RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

HOST INTERFACE (to be integrated with the bare MCU / Pi header)
SPI0: BCM11=SCK, BCM9=MISO, BCM10=MOSI; BCM8=CE0, BCM7=CE1. INT pins: verify against Pi configuration.
These named nets are LOCAL to this zone. All +3V3 / +5V / GND / CAN backbone nets need hierarchy ports before full-board integration.

Pi connector orientation, self-test and indicator circuits remain to be completed.
PRELIMINARY — incomplete board; not for fabrication

Team 4

Sheet: /Cockpit zone/
File: Cockpit.kicad_sch

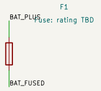
Title: Cockpit zone — dual CAN

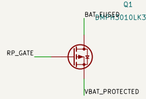
Size: A2	Date: 2026-10-18	**Rev: 0.1**
KiCad E.D.A. 10.0.4		Id: 4/5

AA PACK INPUT — 10x NiMH recommended; exact cells/holder not selected

Design input envelope: 8–15 V at the protected node; verify real cells, contact resistance and sag.
Battery / motor rail is UNREGULATED. No lithium charger or lithium BMS is included.











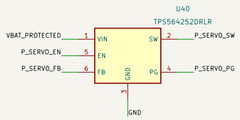






Q1 loss at 10 A: 1 W using 10 mOhm at 25 C; hotter silicon has higher resistance.
F1/holder/cable limits, transient overvoltage and thermal layout are open release checks.
J40 generic footprint is a visible placeholder; DO NOT use a small signal header for battery current.

SERVO: 6.6 V; allocation 4 A; TPS564252 (4 A device rating)

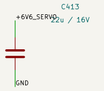




















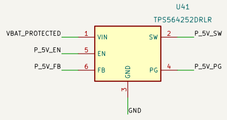






Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

5V: 5.1 V; allocation 3.3 A; TPS564252 (4 A device rating)





















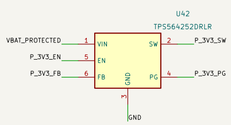






Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

3V3: 3.318 V; allocation 0.8 A; TPS564252 (4 A device rating)



























Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

Calculated component choices; load, thermal and protection validation outstanding.
PRELIMINARY — incomplete board; not for fabrication

Team 4

Sheet: /Power/
File: Power.kicad_sch

Title: Power — AA input and three buck converters

Size: A2	Date: 2026-10-20	**Rev: 0.2**
KiCad E.D.A. 10.0.4		Id: 5/5